

Association between physical activity, body composition, and cognitive performance among female office workers

Yu-Lun Chen^{a,b,*}, Che-Chien Chang^{b,c}, Meng-Ping Lin^{a,b},
Chia-Chi Lin^a, Pei-Yu Chen^a, and Chi-Hung Juan^{a,b}

^aInstitute of Cognitive Neuroscience, National Central University, Taoyuan City, Taiwan

^bCognitive Intelligence and Precision Healthcare Research Center, National Central University, Taoyuan City, Taiwan

^cOffice of Physical Education, National Central University, Taoyuan City, Taiwan

*Corresponding author: e-mail address: yulunchen1991@gmail.com

Abstract

Regular physical activity can potentially prevent cognitive decline. While most studies focused on the general decline of the elderly and child and adolescent population, aging is a gradual process and cognitive decline can commence in middle age. Other than the middle-aged working population, gender-specific nuances are another overlooked area regarding the relationship between physical activity and cognitive performance.

Therefore, this study examines the associations and benefits of maintaining regular physical activity habits with cognitive function and body composition in middle-aged female office workers. The results show that middle-aged females exhibited age-related declines in working memory, while no significant age-related changes are observed in reaction time and executive function. However, the regular exercise group demonstrates the ability to maintain their cognitive performance across age, unlike the sedentary group, who experiences declines in reaction time and executive function with age.

Our findings highlight the significant impact of age on specific cognitive functions in middle-aged females and the positive influence of regular exercise on cognitive performance. Furthermore, this study demonstrates the potential of “the Brain Gym” App for efficient cognitive function assessment. The findings underscore the importance of regular exercise for cognitive well-being in middle-aged females and provide valuable insights into the relationship between body composition and cognitive function.

Keywords: Regular exercise, Body composition, Cognitive function, Middle-aged female



1. Introduction

Several studies have shown that the most critical changes in cognition with normal aging are declines in performance on cognitive tasks that require one to quickly process or transform information to make a decision, including measures of speed of processing, working memory, and executive cognitive function (Park et al., 2002). This decline is typically gradual, starting in midlife and becoming more pronounced in later years (Salthouse, 2012). While these changes are considered a part of the natural aging process, it is essential to distinguish between age-related decline and more severe cognitive impairments.

Although previous studies primarily focused on the elderly population. Nonetheless, emerging evidence suggests that cognitive decline can commence earlier in life, particularly during middle age, generally defined as the age range from 40 to 65 (Salthouse, 2019). Cognitive functions affected during middle age include memory, processing speed, and executive functions (Ferreira et al., 2017). Understanding cognitive decline during this life stage is critical, as it has implications for the individual's quality of life and ability to perform at work and engage in daily activities. In light of this, some individuals in the middle-aged group may maintain cognitive performance, while others exhibit gradual cognitive decline. Nevertheless, some middle-aged individuals may experience more rapid cognitive deterioration (Rönnlund and Nilsson, 2006). Recognizing these varying trajectories is essential for developing targeted interventions and providing appropriate support.

Regular physical activity improves both physical and mental functionality, delays the onset of chronic diseases, and enables older individuals to maintain their mobility and quality of life (e.g., McPhee et al., 2016; Tsukita et al., 2022). Previous studies have also demonstrated that regular physical activity selectively enhances cognitive functions such as working memory, inhibitory control, and executive function both in young and elderly groups (Kamijo and Takeda, 2009, 2010; Themanson and Hillman, 2006; Wang and Tsai, 2016; Wang et al., 2014). However, the impact of physical activity on cognitive function may depend on the nature of specific cognitive tasks and their execution. Therefore, investigating the moderating role of regular physical activity on cognitive functions is crucial for maintaining overall physical health. Moreover, physical activity gradually increases from childhood to adulthood, reaching its peak during adulthood

before gradually declining, and this decline accelerates with age (McPhee et al., 2016). Previous studies on the relationship between cognitive function and physical activity have revealed that investigating the correlation between physical activity levels and cognitive function during middle age can help us better understand and promote the importance of regular physical activity among this age group. Furthermore, previous studies have predominantly focused on children in development or older adults (Huang et al., 2023; Voss et al., 2016; Xue et al., 2019; Zheng et al., 2022), leaving limited discussions on the impact of regular exercise on cognitive function in middle-aged groups (Cox et al., 2016).

On the other hand, sedentary work behaviors negatively impact the physiological functioning and health of the middle-aged population (Falck et al., 2017). For office workers, who often engage in sedentary work, integrating physical activity into their routines has been shown to positively impact cognitive abilities, including memory, attention, and executive function (Chang et al., 2012). However, gender-specific nuances in the relationship between physical activity and cognitive performance remain underexplored. Thus, this study is expected to clarify the associations and benefits of regular physical activity habits among middle-aged female office workers on cognitive performance. In addition, body composition, including the ratio of fat and muscle mass, has gained recognition as an influential factor in cognitive function (Baumgartner, 2000; Sui et al., 2020). Recent research suggests that body composition indicators, such as muscle mass, may correlate with cognitive abilities (Crespillo-Jurado et al., 2019; Li et al., 2023). Additionally, studies have found a potential role of body mass index (BMI) in cognitive performance, although the results are inconsistent (cf. Narciso et al., 2019). Understanding how specific body composition factors may interact with cognitive performance is essential, especially within the office worker population.

Thus, this study sets out to examine middle-aged female office workers to clarify the associations and benefits of maintaining regular physical activity habits with cognitive function and overall physical health. Additionally, we explore the correlation between specific body composition indicators and cognitive performance, thereby linking key indicators that influence cognitive function performance through physical activity. Therefore, we hypothesize that particular body composition indicators arising from regular physical activity can reveal a connection between regular exercise and cognitive performance, demonstrating significant correlations with specific cognitive function outcomes.



2. Methods

2.1 Participants

Ninety-three middle-aged female staffs (age range: 35–63 years) were recruited from National Central University, Taiwan. Fifty of them have regular exercise habits at least three times per week (Aerobic exercise, Running, Fitness, and Open-skilled sports; Average age = 43.6 years, SD = 6.6 years). The remaining 43 female staffs were sedentary and had no regular sport/exercise habits (Average age = 45.6 years, SD = 7.4 years). No individuals reported having a history of neurological problems or cardiovascular diseases, nor were any taking medications that affect cognitive functions, and all participants had normal or corrected-to-normal visual acuity. The present study was conducted with the approval of the Social and Behavioral Research Ethical Principles and Regulations of National Taiwan University, Taiwan.

2.2 Apparatus

2.2.1 Cognitive function APP measurement

The cognitive tests were developed as an application, “The Brain Gym,” compatible with the Apple iOS platform and installed on seventh-generation Apple iPads. These tests were divided into three tasks, each assessing performance in cognitive functions associated with simple reaction time and movement reaction time, working memory, and executive functions, as outlined below.

2.2.1.1 Simple reaction and movement time

The simple reaction task primarily measures reaction time during the preparation phase and performance in response speed for movement (see Fig. 1). Each trial begins by placing a finger on the red dot, and when the blue dot appears, the participants are instructed to move and tap the blue dot with the same finger as quickly as possible. Participants were instructed to use the dominant hand, and finger holder (the bottom of the screen on the non-dominant side) of the non-dominant index finger during the process. There are five trials for practice before the formal trial. There are 10 trials in this task.

2.2.1.2 Working memory

The sequential memory task evaluated the participants’ working memory (see Fig. 2). The task instructed participants to memorize the sequence in

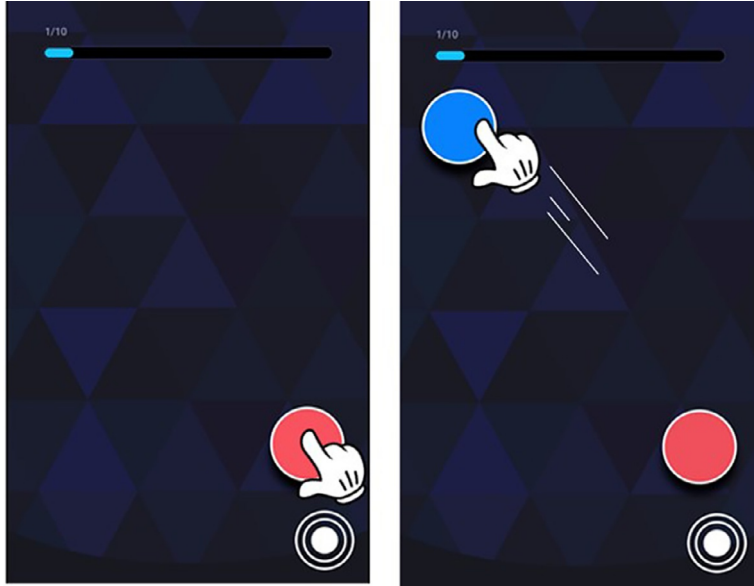


Fig. 1 The simple reaction task in the Brain Gym App. Participants begin by touching and holding onto an object (represented as a red dot) to initiate the trial. Following a random interval of 3–5 s, a target object (depicted as a blue dot) will appear. Participants were instructed to raise their index finger of their dominant hand promptly and tap the target object as swiftly as possible.

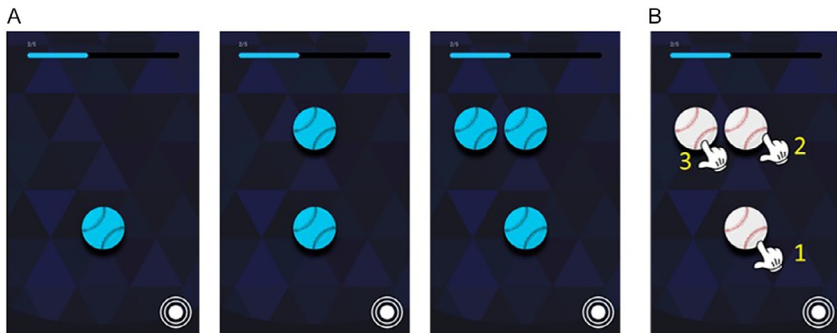


Fig. 2 The sequential memory task in the Brain Gym App. Before the formal session, five interactive trials were conducted to familiarize participants with the task procedure and allow for practice. (A) Initially, dimmed ball stimuli were presented one at a time in random locations on the screen. The inter-stimulus interval (ISI) was defined as 250 ms. Participants were tasked with memorizing these ball stimuli's location and presentation sequence. (B) When the ball became illuminated, participants had to tap each ball in the order they were initially presented. The set size gradually increased over a series of trials, ranging from a minimum of four targets to a maximum of six. Participants were instructed to respond with both accuracy and speed.

which the dimmed ball stimuli were appeared on the screen. Once the balls became illuminated, participants were required click on them in the order they appeared. Same as the previous task, participants were instructed to use the dominant hand, and finger holder (the bottom of the screen on the non-dominant side) of the non-dominant index finger during the process. The task begins with a practice round of three balls stimulus, totaling five trials to familiarize the participants with the game mechanics. Participants must respond correctly to at least four trials to proceed to the formal trial.

Any incorrect trial or responses that took over 2s will automatically lead to the subsequent trial. The task comprises 45 trials, divided into 3 blocks with 4, 5, and 6 balls of stimulus appearing sequentially, with 15 trials in each block. This task assesses response accuracy, memory capacity (how many balls can be remembered), and response times (the time between clicking on the balls).

2.2.1.3 Executive function

The executive planning task measured the executive function in middle-aged female groups. The task primarily gauges short-term memory and executive functioning capabilities (see Fig. 3). Participants were instructed

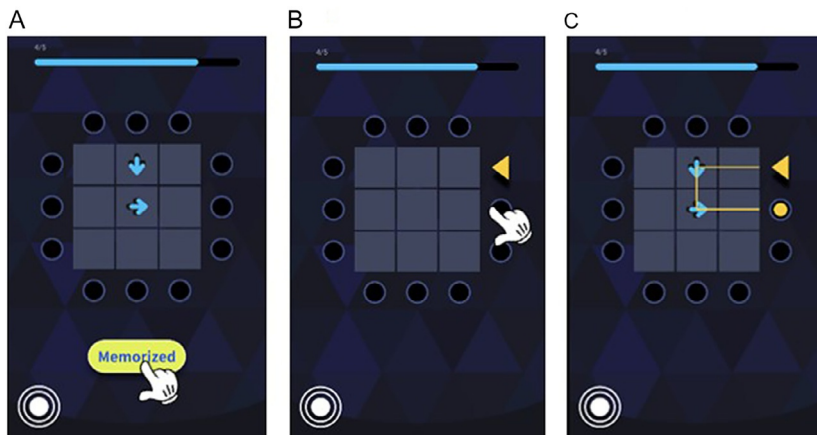


Fig. 3 The executive planning task in the Brain Gym App. (A) At the onset of each trial, participants were presented with sets of 2, 3, and 4 arrows, requiring them to memorize the direction indicated by the arrowheads. The arrows disappeared after pressing the “Memorized” button, then a yellow triangle appeared. (B) The correct path involved connecting the arrow within the grid indicated by the yellow triangle to the corresponding dot along its trajectory. (C) If a participant selected the wrong answer, the chosen dot turned red, and the correct answer was displayed along with the accurate path as a feedback. To succeed in the task, participants must manage and utilize multiple pieces of spatial information, including item positions and arrow directions.

to use the dominant hand, and finger holder (the bottom of the screen on the non-dominant side) of the non-dominant index finger during the process. Each trial starts with 2–4 arrow symbols appearing within a 9-cell grid on the screen. Participants were instructed to remember the position of each arrow and the direction to which it points. After pressing the start answer button (displaying “Memorized”), the arrows disappear, and a starting point of a triangle appears at 1 of the 12 positions surrounding the grid simultaneously. Participants were instructed to respond to the path’s endpoint based on the direction of the disappeared arrows. This task consists of 60 trials, divided into 3 blocks with 2, 3, and 4 arrow conditions appearing sequentially, totaling 20 trials in each block. The task allows for assessing response accuracy, memory capacity (how many arrows can be remembered), and planning capacity.

2.2.2 Body composition

Body composition was assessed using a bioelectrical impedance analysis (BIA) technique at frequencies of 6.25 and 50 kHz, with a current of 90 μ A. The TANITA MC-980 MA PLUS (Tanita, Tokyo, Japan) was employed for these measurements, featuring eight electrodes, with four integrated into the platform and the remainder located in the handles. Similar equipment was applied in previous studies investigating the effect of sarcopenia-related parameters on disability onset in the elderly group (e.g., [Morikawa et al., 2023](#); [Toyoda et al., 2019](#)). Participants stood upright on the platform with their legs extended, ensuring their feet made contact with both front and rear electrodes, distributing their weight evenly across both feet. The body composition parameters measured included BMI (kg/m^2), BMR (kcal), fat (%), fat mass (kg/m^2), trunk fat mass (kg/m^2), muscle mass (kg), and fat-free mass (kg/m^2).

2.3 Procedure

All participants engaged in a two-phase experimental procedure: the initial phase involved body composition measurement, while the second phase consisted of the cognitive tasks. The entire experimental process lasted approximately 30 min.

2.4 Data analysis

The statistical analysis was completed with SPSS version 21.0 software (SPSS, Inc., Chicago, IL) to analyze demographic and cognitive behavioral

results. All participants' descriptive characteristics (including age, height, weight, BMI, BMR, fat, fat mass, muscle mass, trunk fat mass, and fat-free mass) were described using means and standard deviations, and group differences were analyzed using a one-way analysis of variance (ANOVA) method. A behavioral analysis (Pearson correlation) was then conducted to examine the correlation between cognitive performance and body composition and whether the regular exercise group showed enhanced cognitive performance. The level of significance was set at $P \leq 0.05$.



3. Results

3.1 Participant demographics

Participant demographic data and their variables of body composition are provided in Table 1. Demographic variables including age, height, weight, BMI, BMR, fat, fat mass, trunk fat mass, and muscle mass, did not differ between groups (age, [$F(1, 91) = 1.73, P = 0.192, \eta^2 = 0.019$]; height, [$F(1, 91) = 1.02, P = 0.316, \eta^2 = 0.011$]; weight, [$F(1, 91) = 0.005, P = 0.945, \eta^2 = 0.000$]; BMI, [$F(1, 91) = 0.150, P = 0.699, \eta^2 = 0.002$]; BMR, [$F(1, 91) = 1.26, P = 0.264, \eta^2 = 0.014$]; fat, [$F(1, 91) = 2.89,$

Table 1 Demographics of participants in each group.

Group	Regular exercise group ($n = 50$)		Sedentary group ($n = 43$)	
	Mean	SD	Mean	SD
Age (years)	43.6	6.6	45.6	7.4
Height (cm)	160.2	5.2	159.1	4.9
Weight (kg)	58.5	9.3	58.6	10.7
BMI (kg/m^2)	22.8	3.5	23.1	3.5
BMR (kcal)	1170.8	114	1141.1	141.1
Fat (%)	32.6	6.9	33.9	6.3
Fat mass (kg/m^2)	19	7.2	20.4	7.4
Trunk fat mass (kg/m^2)	10.2	4.7	11.4	4.6
Muscle mass (kg)	37.2	2.9	36	3.6
Fat-free mass (kg/m^2)	42.4*	7.6	39*	5.3

SD = standard deviation; BMI = body mass index; BMR = basal metabolic rate. * $P < 0.05$.

$P=0.093$, $\eta^2=0.031$]; fat mass, [$F(1, 91)=0.90$, $P=0.345$, $\eta^2=0.010$]; trunk fat mass, [$F(1, 91)=1.65$, $P=0.202$, $\eta^2=0.018$]; and muscle mass, [$F(1, 91)=3.09$, $P=0.082$, $\eta^2=0.033$]. In contrast, there was a significant group difference for the fat-free mass, [$F(1, 91)=5.88$, $P=0.017$, $\eta^2=0.061$].

3.2 Age and cognitive performance

3.2.1 Simple reaction time

In this analysis, we aimed to examine whether age will affect the reaction performance in exercise habits. The results show that there is a noticeable correlation between age and reaction time in the overall context for the middle-aged female group when comparing those with and without exercise habits (see Fig. 4), the results exhibited that the reaction performance of the

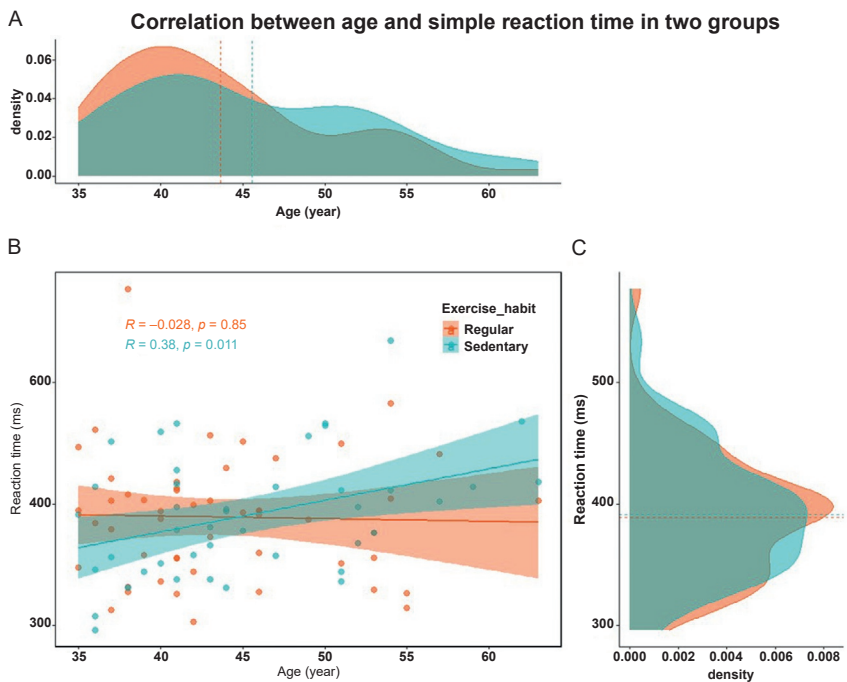


Fig. 4 The correlation between age and simple reaction time. (A) The distribution of age between the regular exercise group and the sedentary group. (B) The correlation between the simple reaction time and age in the two groups. (C) The distribution of simple reaction time performance between the regular exercise and the sedentary group, there is no significant difference between two groups.

sedentary group showed a significant positive correlation ($r = 0.38$, $P = 0.011$), indicating that the reaction time tends to increase could be affected with age. In contrast, the regular exercise group can maintain their reaction performance across age ($r = -0.028$, $P = 0.85$).

3.2.2 Working memory

To investigate the effects of age and working memory abilities in the middle-aged female group, the results showed a significant negative correlation ($r = -0.214$, $P = 0.040$) between age and working memory performance, indicating that working memory performance declines with age in the middle-aged female group (see Fig. 5). Furthermore, no significant correlation was found between age and working memory performance within the specific group effect.

3.2.3 Executive function

Fig. 6 shows the correlation between age and executive planning performance in different groups. The results indicate that among the sedentary group, there was a significant decline in the performance of executive functions with increasing age ($r = -0.31$, $P = 0.041$). In contrast, the regular exercise group maintained similar performance levels as they aged ($r = -0.086$, $P = 0.55$).

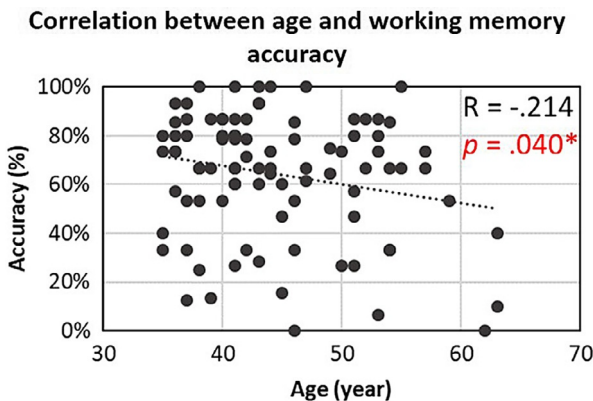


Fig. 5 The correlation between age and working memory performance in the middle-aged female group. The results showed a significant negative correlation ($P < 0.05$), indicating that the working memory capacities of the middle-aged female group could decrease during the age increase in the two groups.

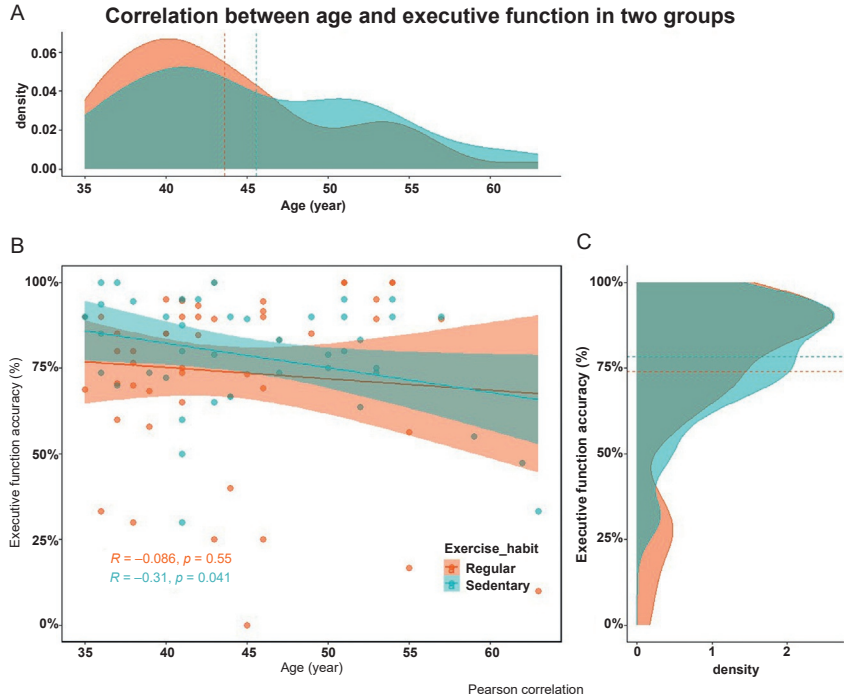


Fig. 6 The correlation between age and executive function. (A) The distribution of age between the regular exercise group and the sedentary group. (B) The correlation between the executive planning performance and age in the two groups. A significant difference between the regular exercise group and the sedentary group was found. The regular exercise group could maintain the executive function performance along the increase of age, conversely, the sedentary group exhibited the cognitive decline with age. These results potentially reveal that regular exercise can positively affect executive function. (C) The distribution of executive planning performance comparison between the regular exercise and the sedentary group, there is no significant difference between two groups.

3.3 Cognitive performance and body composition

In addition to analyzing age and cognitive function changes due to exercise habits, we further investigated whether body composition results could reflect cognitive performance. Table 2 shows the results of Pearson's correlation analysis. This analysis indicated the relationships between body composition and cognitive performance. For the overall middle-aged female

Table 2 Pearson correlation coefficients overall for combined body composition and cognitive performance.

Group	Overall (<i>n</i> = 93)						
	RT	WM			EF		
		SS4	SS5	SS6	SS4	SS5	SS6
BMI (kg/m ²)	−0.008	0.20	0.087	−0.006	−0.17	−0.10	−0.038
BMR (kcal)	−0.089	0.21*	0.10	0.048	−0.10	−0.005	0.14
Fat (%)	0.004	0.11	0.006	−0.066	−0.15	−0.099	−0.034
Fat mass (kg/m ²)	0.009	0.17	0.055	−0.025	−0.13	−0.061	0.001
Trunk fat mass (kg/m ²)	0.031	0.14	0.03	−0.041	−0.14	−0.082	−0.013
Muscle mass (kg)	−0.142	0.22*	0.10	0.064	−0.09	0.022	0.18
Fat-free mass (kg/m ²)	−0.065	0.11	0.053	0.044	−0.05	−0.097	0.13

RT = reaction time; WM = working memory; EF = executive function; SS4 = set size 4; SS5 = set size 5; SS6 = set size 6; BMI = body mass index; BMR = basal metabolic rate. * $P < 0.05$.

group, the results show that working memory performance positively correlates with BMR ($r = 0.21$, $P = 0.039$) and muscle mass ($r = 0.22$, $P = 0.036$). These results indicate that the BMR and muscle mass index may reflect the lower working memory performance.

Table 3 shows the results of Pearson's correlation analysis when comparing the two groups, and it was observed that the regular exercise group exhibited a negative correlation between executive function and BMI ($r = -0.29$, $P = 0.044$). Indicating that the higher BMI impacted the executive function performance in the regular exercise group. Furthermore, in the sedentary group, it was found that higher executive function status was significantly positively correlated with BMR ($r = 0.36$, $P = 0.016$), suggesting that higher BMR rates were associated with better executive function in the absence of regular exercise. Interestingly, similar to the BMR rates, the results also indicated a significant positive correlation between higher executive function status and muscle mass ($r = 0.38$, $P = 0.011$) in the sedentary group, implying that greater muscle mass was associated with improved executive function without a regular exercise habit.

Table 3 Pearson correlation coefficients by group for combined body composition and cognitive performance.

Group	Regular exercise group (n = 50)								Sedentary group (n = 43)						
	RT	WM			EF				RT	WM			EF		
		SS4	SS5	SS6	SS2	SS3	SS4	SS4		SS5	SS6	SS2	SS3	SS4	
BMI (kg/m ²)	−0.14	0.16	0.031	−0.17	−0.17	−0.17	−0.29*	−0.059	0.25	0.13	0.14	−0.16	−0.005	0.26	
BMR (kcal)	−0.019	0.23	0.15	0.033	−0.066	−0.004	−0.07	−0.19	0.20	0.078	0.075	−0.10	0.023	0.36*	
Fat (%)	−0.11	0.14	−0.006	−0.22	−0.12	−0.12	−0.24	0.031	0.11	0.011	0.097	−0.15	−0.058	0.22	
Fat mass (kg/m ²)	−0.12	0.20	0.041	−0.16	−0.12	−0.091	−0.24	−0.037	0.18	0.061	0.13	−0.14	−0.025	0.30	
Trunk fat mass (kg/m ²)	−0.095	0.17	0.003	−0.20	−0.11	−0.11	−0.24	−0.028	0.15	0.062	0.15	−0.14	−0.049	0.28	
Muscle mass (kg)	0.012	0.23	0.19	0.11	−0.028	0.037	0.002	−0.22	0.21	0.052	0.045	−0.12	0.031	0.38*	
Fat-free mass (kg/m ²)	−0.008	0.063	0.033	0.16	−0.12	−0.15	0.014	−0.24	0.14	0.051	−0.073	0.003	−0.026	0.26	

RT = reaction time; WM = working memory; EF = executive function; SS4 = set size 4; SS5 = set size 5; SS6 = set size 6; BMI = body mass index; BMR = basal metabolic rate.

**P* < 0.05.



4. Discussion

The present study investigated the relationship between the age and cognitive performance of middle-aged female office workers. Additionally, it explores whether regular physical activity habits impact cognitive functions. This study also further examines whether the specific body composition indicators can provide crucial factors in the moderating effects on cognitive functions. We hypothesize that the regular exercise group, as age increases, can maintain better cognitive performance. Furthermore, we expect that the specific body composition indicators resulting from regular physical activity can reveal an association between regular exercise and cognitive performance, showing significant correlations with specific cognitive function performance.

The key findings of this study can be divided into two parts. First, within the entire middle-aged female population, it was observed that working memory performance declines with age. In contrast, this demographic group had no significant age-related decline in reaction time and executive function performance. These findings are partially consistent with previous studies (Alatorre-Cruz et al., 2018; D'Antuono et al., 2022; Klencklen et al., 2017; Park et al., 2002). D'Antuono et al. (2022) indicated that age is negatively correlated with performance on visuo-spatial working memory, which begins to decline gradually after the age of 36. Conversely, the extent of active processing engaged in the task is a crucial determinant of spatial working memory performance (Cornoldi and Vecchi, 2004), and these skills are notably susceptible to the effects of aging (Johnson et al., 2010; Mammarella et al., 2013). Our study extends these findings, revealing that the working memory performance of the middle-aged population continues to decline with age. Therefore, it is crucial to investigate the cognitive performance of middle-aged groups. Furthermore, Brown (2016) investigated the spatial-sequential working memory performance in younger (18–40 years) and older (64–85 years) adults, and the results revealed that age significantly predicted spatial-sequential working memory performance within both the younger and older adult groups. Thus, except for young people and the elderly, our results provide a decline in sequential memory processes in the middle-aged group and also reveal a pronounced trend of rapid decline within this middle-aged stage.

On the other hand, previous studies indicated that physical activity might modify cognitive function in earlier adulthood, and that these changes might

enhance existing cognitive function or attenuate the decline with aging (Barnes, 2015; Reas et al., 2019; Xu et al., 2023). Although this study did not find differences in working memory between the regular exercise group and the sedentary group, however, distinct results were observed regarding response time and executive function performance. We found that the sedentary group exhibited longer reaction time with age, while the regular exercise group maintained their fair level of reaction time performance. The results can be explained as regular exercise maintaining better response performance, while the sedentary group's response performance significantly declines with age. Wolkorte et al. (2014) investigated differences in response preparation in young and middle-aged adults, and the results indicated that the middle-aged adults responded slower than young adults. Interestingly, the results also exhibited a differential effect of age on male and female adults. In middle-aged female participants' reaction times increased more than male ones. In addition, Formenti et al. (2020) examined the effect of exercise on cognitive performance in middle-aged adults, and the results stated that both aerobic and balance exercise were able to positively improve simple reaction time performance in middle-aged adults. Thus, in line with previous research, our results further demonstrate the positive benefits of a regular exercise for female in the middle-age group.

Furthermore, executive function encompasses the regulation and control of cognitive processes (Diamond, 2013; Jurado and Rosselli, 2007), our results indicated that the sedentary group showed a notable decrease in executive function performance with age, while the regular exercise group maintained their executive function performance. Previous studies has shown that executive functions exhibit diverse patterns of development and aging, with continued maturation through early adulthood and subsequent decline in older age (Ferguson et al., 2021; Veríssimo et al., 2022). Moreover, previous studies also revealed that the regular exercise intervention have positive effects on the executive function of the elderly people (Barcelos et al., 2015; Vaughan et al., 2014; Voss et al., 2016; Zheng et al., 2022). Therefore, modest variation in executive function may be due to high levels of functioning at this age stage. As cognitive impairment due to aging typically does not manifest until age 60, positive effects of physical activity may be more difficult to observe in younger healthy populations (Cox et al., 2016). In comparison to previous studies, our results provide further evidence that exercise has positive effects on executive function during middle-age in female adults. As stated above, this study has discovered that age impacts specific cognitive function performance in middle-aged

female, however, having a regular exercise routine can delay the extent of specific cognitive function decline.

Second, we further examine the relationship between body composition and cognitive function in middle-aged females. Previous studies have primarily examined the relationship between cognitive function and indicators such as BMI and fat related indexes (Huang et al., 2019; Kumpulainen et al., 2016; Noh et al., 2017; Seo et al., 2021; Vints et al., 2023). Currently studies demonstrate that older adults had higher BMI due to increased fatness, reduced cardiorespiratory and metabolic function and performed worse in cognitive tests than young ones (Bijlsma et al., 2013; Sillanpää et al., 2008). In addition, Narimani et al. (2019) investigated the association between executive function and weight status in young adults. The results provided that high fatness is associated with worse inhibitory control. Interestingly, Huang et al. (2019) revealed that higher fat mass was also associated with worse performance on measures of cognitive control in young adults.

Furthermore, previous studies have largely focused on children and the elderly, while our study expands its scope to include the middle-aged groups and examines a broader range of body composition indicators. In this study, we found that a positive correlation between lower working memory performance and BMR in middle-aged females. Similarly, there is a positive correlation between muscle mass and working memory performance. These results suggest that BMR and muscle mass may be associated with better performance in tasks requiring a lower working memory load. Conversely, no significant correlation was found in situations requiring a higher working memory load.

On the other hand, when the middle-aged female adults were divided into groups based on regular exercise and sedentary, it was observed that there was a positive correlation between higher-order executive function performance and BMR in the sedentary group. The same results were also found in muscle mass. Thus, these results reveal that the higher BMR and muscle mass can be associated with better executive function performance in the sedentary group. Interestingly, given these participants were in the moderate range of BMI (range: 18–26), a negative correlation was found between BMI and higher-order executive function performance in the regular exercise group. This result indicates that a higher BMI may decrease executive function performance, even with regular exercise habits.

The present study suggests that the aging process in middle-aged females is associated with cognitive function variations, with regular exercise

showing positive benefits for executive function and response time performance in this group. Thus, for those who were relatively sedentary through middle age, to beginning a regular exercise regimen leads to significant improvements to health (Berk et al., 2006; Hamer et al., 2014; Sabia et al., 2012; Sun et al., 2010) and cognition (Lautenschlager et al., 2008). In addition, high cardiovascular fitness may be particularly beneficial for brain volume in prefrontal regions, which involves executive functions, decision making, learning and memory (Miller and Cohen, 2001), consistent with the selective effects of chronic exercise on executive function shown in recent studies (Chen et al., 2020; Huang et al., 2023; Liu et al., 2020; Xue et al., 2019). Furthermore, this study demonstrates that measurements through our cognitive task in the APP can effectively measure these effects, offering a more convenient and efficient way of assessing cognitive function changes compared to conventional tools. It is important to note that this research is limited to middle-aged female office workers, and future studies need to extend to different types of profession in the middle-aged female group.



5. Conclusion

In summary, this study investigates the complex interplay among age, cognitive performance, physical activity, and body composition within the context of middle-aged female office workers. Our hypotheses posit that regular exercise may contribute to the preservation of cognitive performance as middle-aged women age, and this link between regular exercise and cognitive performance may be linked to specific body composition indicators influenced by physical activity.

The present study suggests that age significantly impacts specific cognitive function performance in middle-aged females. However, our findings suggest that the potential for regular exercise to mitigate the extent of cognitive decline in this population. Furthermore, the study demonstrates the efficacy of mobile devices in measuring cognitive function changes conveniently and efficiently, offering a promising alternative to traditional assessment tools.

Conflict of interest

We declare no competing financial interests.

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